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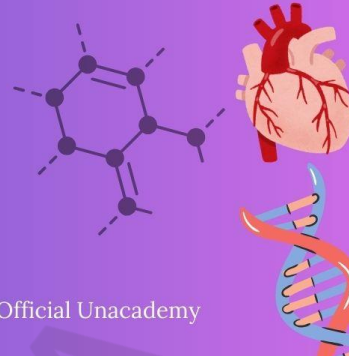
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LAST 10 YEARS

NEET PYQ



Chapter 10 : Organisms and Populations

1. Epiphytes that are growing on a mango branch is an example of which of the following?

(1) Commensalism

[NEET-2025]

(2) Mutualism

(3) Predation

(4) Amensalism

2. Which one of the following equations represents the Verhulst-Pearl Logistic Growth of population?

[NEET-2025]

(1) $\frac{dN}{dt} = r \left(\frac{K - N}{K} \right)$

(2) $\frac{dN}{dt} = rN \left(\frac{K - N}{K} \right)$

(3) $\frac{dN}{dt} = rN \left(\frac{N - K}{N} \right)$

(4) $\frac{dN}{dt} = N \left(\frac{r - K}{K} \right)$

3. Given below are two statements :

[NEET-2025]

Statement I : Fig fruit is a non-vegetarian fruit as it has enclosed fig wasps in it.

Statement II : Fig wasp and fig tree exhibit mutual relationship as fig wasp completes its life cycle in fig fruit and fig fruit gets pollinated by fig wasp.

In the light of the above statements, choose the most appropriate answer from the options given below :

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

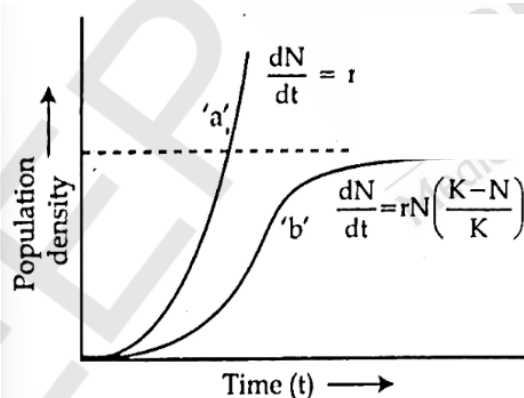
4. Identify When will the population density increase, under the special conditions? When the number of : [Re-NEET-2024]
- (1) Deaths exceeds number of births and also number of emigrants equals number of immigrants.
 - (2) Births plus number of immigrants equals number of deaths plus number of emigrants.
 - (3) Births plus number of emigrants is more than the number of deaths plus number of immigrants.
 - (4) Births plus number of immigrants is more than the sum of number of deaths and number of emigrants.

5. Match List-I with List-II: [Re-NEET-2024]

List I	List II
A. Predator	I. Ophrys
B. Mutualism	II. Pisaster
C. Parasitism	III. Female wasp and fig
D Sexual deceit	IV. Plasmodium

Chose the correct answer from the options given below:

- (1) A-III, B-II, C-I, D-IV
 - (2) A-IV, B-I, C-II, D-III
 - (3) A-II, B-III, C-I, D-IV
 - (4) A-II, B-III, C-IV, D-I
6. What do 'a' and 'b' represent in the following population growth curve? [Re-NEET-2024]



- (1) 'a' represents exponential growth when responses are not limiting the growth; and 'b' represents logistic growth when responses are limiting the growth.
- (2) 'a' represents logistic growth when responses are not limiting the growth; 'b' represents exponential growth when responses are limiting the growth.
- (3) 'a' represents carrying capacity and 'b' shows logistic growth when responses are limiting the growth.
- (4) 'a' represents exponential growth when responses are not limiting the growth and 'b' shows carrying capacity.

7. The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN \left[\frac{K-N}{K} \right]$. From this equation, K indicates: [NEET-2024]

- (1) Intrinsic rate of natural increase (2) Biotic potential
(3) Carrying capacity (4) Population density

8. Given below are two statements: [NEET-2024]

Statement I: Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.

Statement II: According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.

In the light of the above statements, choose the correct answer from the options given below : (1) Both Statement I and Statement II are true.

- (2) Both Statement I and Statement II are false.
(3) Statement I is true but Statement II is false.
(4) Statement I is false but Statement II is true.

9. Plants offer rewards to animals in the form of pollen and nectar and the animals facilitate the pollination process. This is an example of : [NEET-2023-MANIPUR]

- (1) Commensalism (2) Mutualism
(3) Amensalism (4) Competition

10. For chemical defence against herbivores, Calotropis has [NEET-2023-MANIPUR]

- (1) toxic ricin (2) distasteful quinine
(3) cardiac glycosides (4) strychnine

11. If there are 250 snails in a pond, and within a year their number increases to 2500 by reproduction. What should be their birth rate per snail per year? [NEET-2023-MANIPUR]

- (1) 25 (2) 15
(3) 10 (4) 9

12. Given below are two statements: [NEET-2023-MANIPUR]

Assertion (A) : A person goes to high altitude and experiences "Altitude Sickness" with symptoms like breathing difficulty and heart palpitations.

Reason (R) : Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(3) (A) is true but (R) is false
(4) (A) is false but (R) is true

13. Match List I with List II :

[NEET-2023]

List I (Interaction)

List II (Species A and B)

A. Mutualism

I. $+(A), 0(B)$

B. Commensalism

II. $-(A), 0(B)$

C. Amensalism

III. $+(A), -(B)$

D. Parasitism

IV. $+(A), +(B)$

(1) A-IV, B-II, C-I, D-III

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-III, B-I, C-IV, D-II

14. Given below are two statements:

[NEET-2023]

Statement I : Gause's 'Competitive Exclusion Principle' states that two closely related species competing for the same resources cannot co-exist indefinitely and competitively inferior one will be eliminated eventually.

Statement II : In general, carnivores are more adversely affected by competition than herbivores.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is correct but Statement II is false.

(4) Statement I is incorrect but Statement II is true.

15. Match List I with List II.

[NEET-2023]

List I (Interacting species)

List II (Name of interaction)

A. A Leopard and a Lion in a forest/grassland

I. Competition

B. A cuckoo laying egg in a crow's nest

II. Brood parasitism

C. Fungi and root of a higher plant in Mycorrhizae

III. Mutualism

D. A cattle egret and a Cattle in a field

IV. Commensalism

Choose the correct answer from the options given below.

(1) A-I, B-II, C-III, D-IV

(2) A-I, B-II, C-IV, D-III

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-III, C-I, D-IV

16. Match List I with List II.

[NEET-2023]

List I

List II

A. Logistic growth

I. Unlimited resource availability condition

B. Exponential growth

II. Limited resource availability condition

C. Expanding age pyramid followed by

III. The percent individuals of pre-reproductive age is largest reproductive and post reproductive age groups

D. Stable age pyramid age group

IV. The percent individuals of pre-reproductives and reproductive are same

Choose the correct answer from the options given below:

(1) A-II, B-I, C-III, D-IV

(2) A-II, B-III, C-I, D-IV

(3) A-II, B-IV, C-I, D-III

(4) A-II, B-IV, C-III, D-

17. Which one of the following statements cannot be connected to Predation ?

[NEET-2022]

(1) It might lead to extinction of a species

(2) Both the interacting species are negatively impacted

(3) It is necessitated by nature to maintain the ecological balance

(4) It helps in maintaining species diversity in a community

18. While explaining interspecific interaction of population, (+) sign is assigned for beneficial interaction, (-) sign is assigned for detrimental interaction and (0) for neutral interaction. Which of the following interactions can be assigned (+) for one species and (-) for another species involved in the interaction ?

[NEET-2022]

(1) Amensalism

(2) Commensalism

(3) Competition

(4) Predation

19. Two butterfly species are competing for the same nectar of a flower in a garden. To survive and coexist together, they may avoid competition in the same garden by:

[NEET-2022]

(1) feeding at the same time

(2) choosing different foraging patterns

(3) increasing time spent on attacking each other

(4) predating on each other

20. The pioneer species in a hydrarch succession are : [NEET-2022]
(1) Free-floating angiosperms (2) Submerged rooted plants
(3) Phytoplanktons (4) Filamentous algae
21. The species that come to appear in bare area are called : [NEET-2022]
(1) Pioneer species (2) Invasive species
(3) Competitive species (4) Species of seral community
22. All successions irrespective of the habitat proceed to which type of climax community? [NEET-2022]
(1) Xeric (2) Mesic
(3) Hydrophytic (4) Edaphic
23. In spite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival? [NEET-2021]
(1) Resource partitioning (2) Competitive release
(3) Mutualism (4) Predation
24. Amensalism can be represented as : [NEET-2021]
(1) Species A (-); Species B (0) (2) Species A (+); Species B (+)
(3) Species A (-); Species B (-) (4) Species A (+); Species B (0)
25. In the exponential growth equation $N_t = N_0 e^{rt}$, e represents: [NEET-2021]
(1) The base of number logarithms (2) The base of exponential logarithms
(3) The base of natural logarithms (4) The base of geometric logarithms
26. Match List-I with List-II. [NEET-2021]

List-I

List-II

- | | |
|------------------------------|----------------------------|
| (a) Allen's Rule | (i) Kangaroo rat |
| (b) Physiological adaptation | (ii) Desert lizard |
| (c) Behavioural adaptation | (iii) Marine fish at depth |
| (d) Biochemical Adaptation | (iv) Polar seal |

Choose the correct answer from the options given below.

- (1) (a)-(iv), (b)-(ii), (c)-(iii), (d)-(i)
(2) (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
(3) (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
(4) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

27. Assertion (A) : A person goes to high altitude and experiences 'altitude sickness' with symptoms like breathing difficulty and heart palpitations. [NEET-2021]

Reason (R) : Due to low atmospheric pressure at high altitude, the body does not get sufficient oxygen.

In the light of the above statements, choose the correct answer from the options given below.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are true but (R) is not the correct explanation of (A)
- (3) (A) is true but (R) is false
- (4) (A) is false but (R) is true

28. If a pond has 40 lotus plants last year and through reproduction 8 new plants are added, taking the current population to 48, what will be the birth rate? [NEET-2021]

- (1) 0.6 offspring per lotus per year
- (2) 0.5 offspring per lotus per year
- (3) 0.2 offspring per lotus per year
- (4) 0.1 offspring per lotus per year

29. Which of the following is not an attribute of a population? [NEET-2020]

- (1) Species interaction
- (2) Sex ratio
- (3) Natality
- (4) Mortality

30. Match the items in Column-I with those in Column-II : [NEET-2020]

Column I

Column II

- | | |
|-----------------------|------------------|
| (a) Herbivores-Plants | (i) Commensalism |
| (b) Mycorrhiza-Plants | (ii) Mutualism |
| (c) Sheep-Cattle | (iii) Predation |
| (d) Orchid-Tree | (iv) Competition |

- (1) (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
- (2) (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
- (3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)
- (4) (a)-(i), (b)-(iii), (c)-(iv), (d)-(ii)

31. The impact of immigration on population density is :- [NEET-2020]

- (1) Negative
- (2) Both positive and negative
- (3) Neutralized by natality
- (4) Positive

32. Carnivorous animals - lions and leopards, occupy the same niche but lions predate mostly larger animals and leopards take smaller ones. This mechanism of competition is referred to as :-

- | | | |
|----------------------------|---------------------------|-------------|
| (1) Character displacement | (2) Altruism | [NEET-2019] |
| (3) Resource partitioning | (4) Competitive exclusion | |

33. Niche is [NEET-2018]

- (1) all the biological factors in the organism environment
- (2) the physical space where an organism live
- (3) the range of temperature that the organism needs to live
- (4) the functional role played by the organism where it lives

34. Which one of the following plants shows a very close relationship with a species of moth, where none of the two can complete its life cycle without the other? [NEET-2018]

- | | |
|--------------|-----------|
| (1) Hydrilla | (2) Yucca |
| (3) Banana | (4) Viola |

35. Which of the following flowers only once in its life-time ? [NEET-2018]

- | | |
|--------------------|----------------|
| (1) Bamboo species | (2) Jack fruit |
| (3) Mango | (4) Papaya |

36. Natality refers to [NEET-2018]

- (1) Death rate
- (2) Birth rate
- (3) Number of individuals leaving the habitat
- (4) Number of individuals entering a habitat

37. In a growing population of a country [NEET-2018]

- (1) pre-reproductive individuals are more than the reproductive individuals.
- (2) reproductive individuals are less than the post-reproductive individuals.
- (3) reproductive and pre-reproductive individuals are equal in number.
- (4) pre-reproductive individuals are less than the reproductive individuals.

38. Asymptote in a logistic growth curve is obtained when: [NEET-2017]

- (1) $K = N$
- (2) $K > N$
- (3) $K < N$
- (4) The value of 'r' approaches zero

39. Presence of plants arranged into well defined vertical layers depending on their height can be seen best in: [NEET-2017]

- (1) Tropical Rain Forest
- (2) Grassland
- (3) Temperate Forest
- (4) Tropical Savannah

40. Mycorrhiza are the example of : [NEET-2017]

- (1) Amensalism
- (2) Antibiosis
- (3) Mutualism
- (4) Fungistasis

41. In a hypothetical population of 100 individual having 'r' = 0.5/female/year, what will be the population size in 6 years (with e = 2.72) showing exponential rate of growth ? [NEET-2017]

- (1) 1218
- (2) 739
- (3) 2012
- (4) 448

42. When does the growth rate of a population following the logistic model equal zero? The logistic model is given as $\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$:- [NEET-I 2016]

- (1) when N/K is exactly one.
- (2) when N nears the carrying capacity of the habitat.
- (3) when N/K equals zero.
- (4) when death rate is greater than birth rate.

43. Gause's principle of competitive exclusion states that : [NEET-I 2016]

- (1) More abundant species will exclude the less abundant species through competition.
- (2) Competition for the same resources excludes species having different food preferences.
- (3) No two species can occupy the same niche indefinitely for the same limiting resources.
- (4) Larger organisms exclude smaller ones through competition.

44. Which of the following would appear as the pioneer organisms on bare rocks?

- (1) Lichens [NEET-I 2016]
- (2) Liverworts
- (3) Mosses
- (4) Green algae

45. The principle of competitive exclusion was stated by :- [NEET-II 2016]

- (1) MacArthur
- (2) Verhulst and Pearl
- (3) C. Darwin
- (4) G.F. Gause

ANSWER KEY

1. (1)
2. (2)
3. (2)
4. (4)
5. (4)
6. (1)
7. (3)
8. (4)
9. (2)
10. (3)
11. (4)
12. (1)
13. (2)
14. (3)
15. (1)
16. (1)
17. (2)
18. (4)
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36. (2)
37. (1)
38. (1)
39. (1)
40. (3)
41. (3)
42. (1)
43. (3)
44. (1)
45. (4)